

MEHDI SEMMAR

Software Engineer

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Education

University of Ottawa, Ottawa, ON, Canada

Fall 2023 – Fall 2027

BASc Software Engineering (Eng. Manag. & Entrepreneurship) Co-op

- GPA: 4.0/4.0 | x6 Dean's List | TD Green Scholarship | J-D Hearnshaw Scholarship
- Extracurriculars: x4 Hackathon participant | uoTutorat (Math Tutor) | Clubs/Associations: Tennis, Chess, SESA

Work Experience

University of Ottawa

Ottawa, ON, Canada

Research Assistant | Reinforcement Learning for Autonomous Drones (Incoming)

09/2026 – 12/2026

- Assisting Dr. Bellinger on reinforcement learning research applied to autonomous drones.

Vertex Inc.

King of Prussia, PA, United States

Software Engineer Intern | Fintech & Tax Technology

05/2026 – 09/2026

- Designed and built an OpenAPI governance platform of four repositories (a Go engine, Terraform infrastructure, a developer portal, and an audit agent) that enforces API compatibility across all engineering teams.
- Built the core engine as a Go REST service on PostgreSQL, embedding linting (vacuum) and breaking change detection (oasdiff) that reject any spec with errors or a changed contract missing a new semver bump (HTTP 409/422).
- Built an internal developer portal (React/TypeScript, Swagger UI) where engineers browse a detailed view of every Vertex service, its specs, and each version.
- Rolled the platform out through GitHub Actions gates set as required status checks, and added an MCP server that exposes the engine's tools to coding agents.
- Ran an AI agent from a playbook that calls the MCP server's tools (spec diffing, backward compatibility checks) to audit 5 flagship APIs across 1,562 commits and 118 specs, surfacing 1,244 breaking changes across 253 endpoints on frozen versions.
- Won an internal hackathon among 30 engineers with SnapTax, an app that scans a product barcode to return the full price including taxes and the nearby location where it is cheapest.

Raven Connected

Ottawa, ON, Canada

AI/ML Engineer Intern | AI-Driven 3D Reconstruction

09/2025 – 12/2025

- Forecasted leaf-pile volumes from images using SAM3/SAM3D, delivering a model that closed a 4-figure client contract.
- Generated synthetic training data and trained a segmentation model on millions of Raven frames.
- Built an AI pipeline integrating segmentation, VGGT 3D generation, and Gaussian splatting, using Depth Pro height estimation to correct the device's 3D position as a gyroscope calibration substitute.
- Optimized model efficiency and runtime at scale through GPU acceleration and algorithm development (PyTorch, AWS S3).

Versaterm

Ottawa, ON, Canada

Full-Stack Developer Intern | Computer-Aided Dispatch

05/2025 – 08/2025

- Built a desktop app (Electron.js) for VMobile that simplified JSON-based form configuration, giving customer success teams an intuitive interface that improved their efficiency.
- Built and deployed a parser GUI (4JS, Bash, Vim) to standardize data integration rules, letting users create, edit, validate, and test configurations from one interface and cutting rule file writing time by 50%.
- Presented both projects to 20+ engineers and stakeholders to highly positive feedback, delivering production-ready features prepared for client adoption.

Leadership & Mentorship

SESA | Events Team Lead

- Led a 6+ person team delivering technical workshops with partners including Amazon, Ciena, Deloitte, and National Bank of Canada, owning logistics, scheduling, and vendor coordination for seamless delivery.

University of Ottawa | Teaching Assistant, ITI1121 & SEG 2505

- Led weekly labs on OOP, Java, and software engineering fundamentals, grading assignments and giving feedback that improved students' code quality and problem-solving.

Projects

RZO Sports | Startup Garage uOttawa

- Co-founded a two-sided marketplace connecting athletes with sports venues; placed Top 2 of 40+ teams at a pitch competition.
- Leading MVP development with Spring Boot, React.js, MySQL, Docker, Nginx, and JWT/OAuth 2.0 on AWS EC2, architecting the system with scalable design patterns and data-driven features for maintainability and performance.

uOttHack 8 | Web Development

- Built an admin tool for on-site operations, handling hacker check-in and food/meal tracking for 3,000+ users.
- Developed and shipped the official event website, delivering a responsive front end for registration and participant information.

Skills

Languages: Python, R, Java, Go, JavaScript, TypeScript, Dart, C/C++, 4JS, Lua, NASM, SQL, Bash, HTML5, CSS3

Frameworks: TensorFlow, PyTorch, Spring Boot, Flask, Django, Flutter, Vue.js, Electron.js, React.js, Express.js, Node.js

Other: AWS, GCP, Docker, Kubernetes, Terraform, PubSub+, Git, Linux, Neovim, Jupyter Notebook, Postman, Testing, CI/CD